Shared and Distinct Exonic Parts in Alternative Paths of Splicing Bubbles

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INTRODUCTION

- Alternative splicing (AS) is a process that enables different RNA isoforms to be generated from the same gene, expanding the functional capacity of a finite genome.
- Our lab introduced a splicing graph called GrASE (Aquino et al.) that follows the convention as implemented in the SplicingGraph package in Bioconductor, but has been extended to include the exonic part edges along the genome.

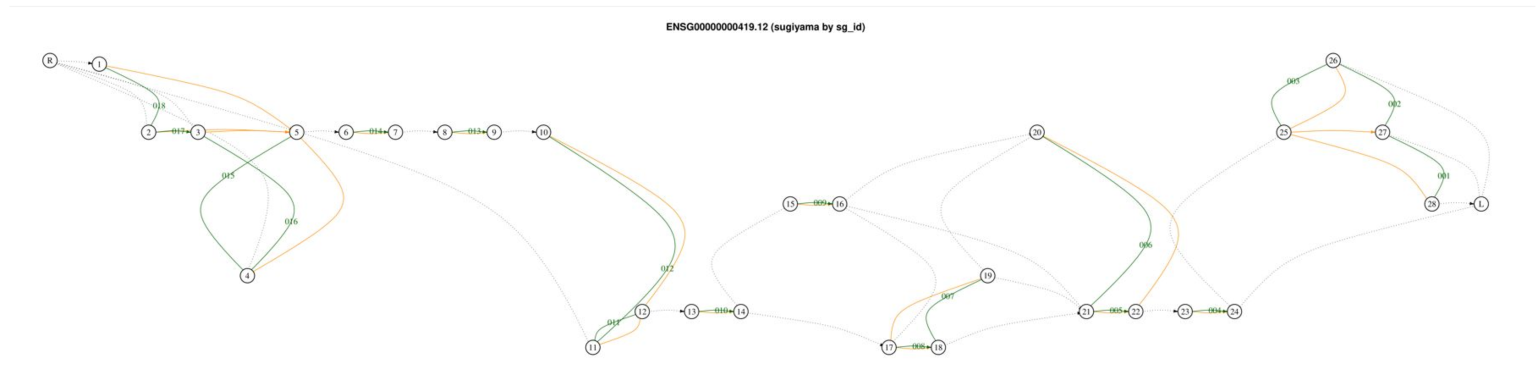


Figure 1. GrASE splicing graph of the DPM1 (ENSG00000000419.12) gene.

- After constructing the splicing graph, we can identify alternative splicing as the complete set of **bubbles (internal/TSSTTS)** based on the gene annotations.
- Valid bubbles are structures in the graph where there exist at least two distinct valid paths – a path followed by at least 1 annotated transcript – between two nodes of a splicing graph.

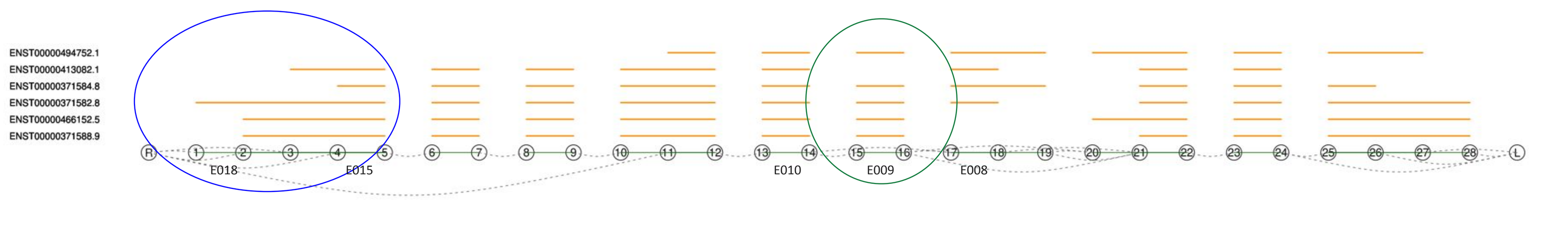


Figure 2. Splicing graph of the six transcript structures of DPM1.

ENSG00000000419.12

- After bubbles are identified, we use **lower-set bipartitioning** to explore possible ways to divide the paths in each bubble into two groups. For each candidate bipartition, we then determine whether any exonic parts distinguish one partition from the other.
- For a valid bipartition, exonic parts can be classified relative to the two partitions: **shared exonic parts** are present in paths from both partitions, while **distinct exonic parts** are present in one partition but absent from the other. These distinct exonic parts are therefore used to distinguish the two partitions after the bipartition has been defined.

source	sink	path	transcripts	shared exonic part	contain distinct exonic part E009
14	17	14-17	ENST00000413082.1	E010,E008	FALSE
14	17	14-15-16-17	ENST00000371582.8, ENST00000371584.8, ENST00000494752.1	E010,E008	TRUE

Table 1. The paths in an internal bubble of DMP1 and their corresponding transcripts, where the shared exonic parts are E010,E008 and the distinct exonic part is E009.

source	sink	path	transcripts	shared exonic part	contain distinct exonic part E018
R	5	R-2-5	ENST00000371588.9, ENST00000466152.5	E015	FALSE
R	5	R-4-5	ENST00000371584.8	E015	FALSE
R	5	R-3-5	ENST00000413082.1	E015	FALSE
R	5	R-1-5	ENST00000371582.8	E015	TRUE

Table 2. The paths in a TSSTTS bubble of DMP1 and their corresponding transcripts, where the shared exonic part is E015 and the distinct exonic part is E018.

STATISTICAL ANALYSIS

- **Research Question: With our defined internal and TSS/TTS splicing bubbles, does cell type A exhibit a higher proportion of reads assigned to the partition containing distinct exonic parts compared with cell type B?**
- **Statistical Model**
  For each bipartition event, let:
  - y = reads assigned to the partition containing distinct exonic parts
  - n = total reads for shared + distinct exonic parts

  To account for overdispersion in read counts, we modeled partition usage using a **beta-binomial generalized linear model**.

- Alternative model (cell-type effect):
  m1: glmmTMB(cbind(y, n − y) ~ Group, family = betabinomial(link = "logit"))
- Null model (no cell-type effect):
  m0: glmmTMB(cbind(y, n − y) ~ 1, family = betabinomial(link = "logit"))

Both models were fitted using a beta-binomial distribution with a logit link.
Models were compared using **a Likelihood Ratio Test (LRT):** LR=2[logLik(m1)−logLik(m0)]

- **Dispersion (ϕ) Estimation Strategies**
  A key challenge is estimating the beta-binomial **dispersion parameter ϕ**, which captures extra variability beyond the binomial model. We compared four strategies:
  1. **Wilcoxon Test:**
     a. Non-parametric baseline
     b. Ignores the beta-binomial distribution entirely
  2. **Beta-binomial MLE:**
     a. Estimates dispersion independently for each event and trusts the event as it is
     b. No prior or shrinkage
  3. **EBmap:**
     a. Empirical Bayes moderation using MAP (Maximum A Posteriori) optimization
     b. A global prior is passed directly to glmmTMB
     c. The model estimates ϕ by balancing event-specific evidence with the empirically determined prior
  4. **EBapprox:**
     a. Empirical Bayes moderation using a closed-form Gaussian approximation
     b. Dispersion is estimated first, then shrunk toward a global prior
     c. The moderated ϕ is fixed in glmmTMB using the map argument

SIMULATION

- To evaluate the performance of statistical models for detecting differential alternative splicing, we conducted a simulation study based on the simulated data from Love et al.
- Genes were classified into four simulation types: genes with no induced change (Background), genes with differential gene expression only (DGE), genes with differential transcript expression (DTE), and genes with differential transcript usage (DTU).
- In the simulation, Background and DGE genes serve as null genes for splicing detection. Background genes have no induced change, while DGE genes have only gene-level expression changes in which all transcripts are scaled together. Because transcript proportions remain unchanged in DGE genes, they should not produce differential transcript usage or differential splicing calls. Therefore, significant calls in Background or DGE genes are treated as false positives. This allows us to evaluate whether each statistical model detects true DTE/DTU signals while controlling false positives from genes without true splicing changes.
- Because the simulation defines ground truth at the transcript level, we translated from transcript to exonic part and assigned exonic parts to positive or negative based on their transcript membership. For DTE genes, exonic parts overlapping the transcript with induced expression change were labeled positive. For DTU, two transcripts swap their expression levels between conditions, with no net change in total gene expression, and exonic parts were labeled positive only if they overlapped one swapped transcript but not the other; exonic parts shared by both or neither swapped transcripts were labeled negative.

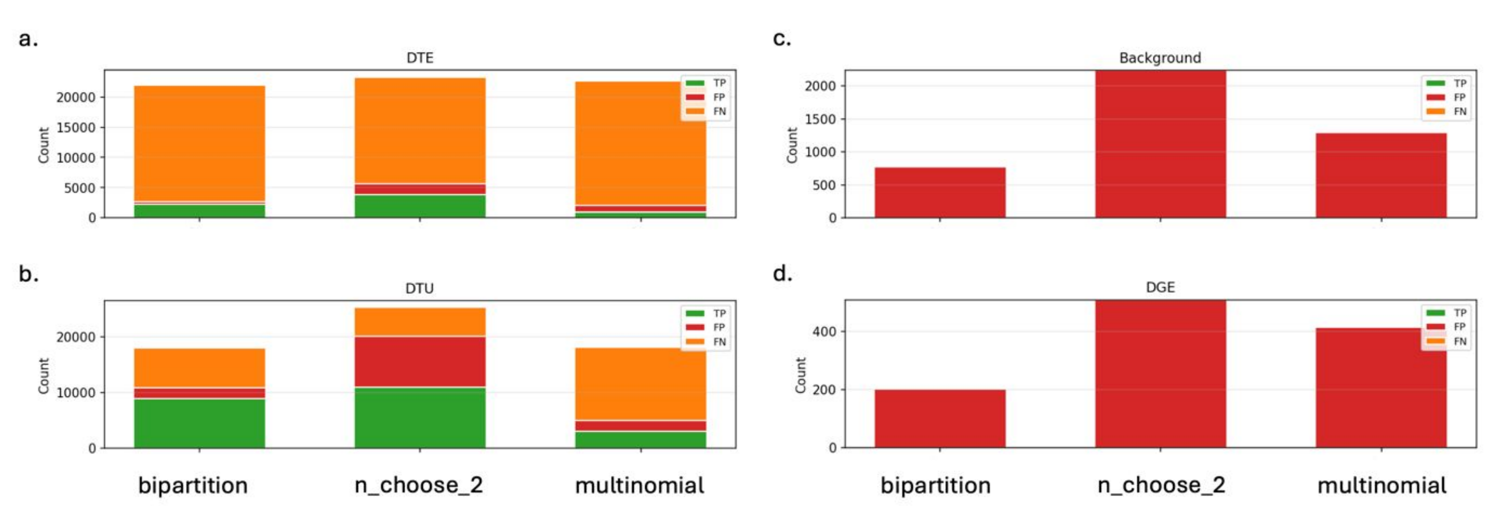


Figure 3. Confusion-count stacked bar charts for three comparison structures at padj ≤ 0.05. Each bar shows the total number of true positives (TP, green), false positives (FP, red), and false negatives (FN, orange) for bipartition, C(k,2), and multinomial comparison structures across simulation categories: DTE, DTU, Background, and DGE. All three comparison structures identify shared and distinct exonic parts so that read coverage can be compared across alternative paths in a splicing bubble. The C(k,2) approach compares every pair of individual paths, while the multinomial approach treats each path as a separate outcome and requires path-unique exonic parts. The result shows that bipartition has the fewest FPs from Background and DGE, indicating well-controlled type I error. C(k,2) generates about five times more false positives from Background and DGE than bipartition or multinomial, reflecting its elevated false positive rate. Multinomial has the largest false-negative counts due to lower test coverage.

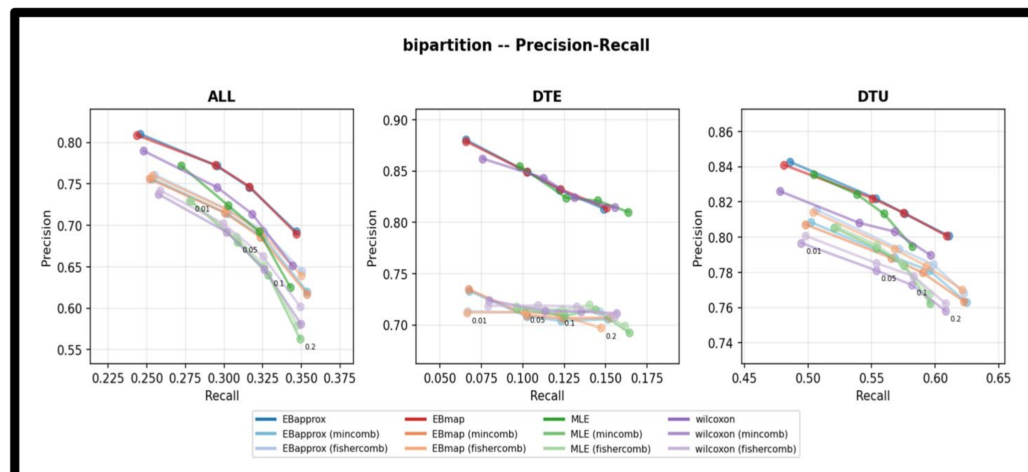
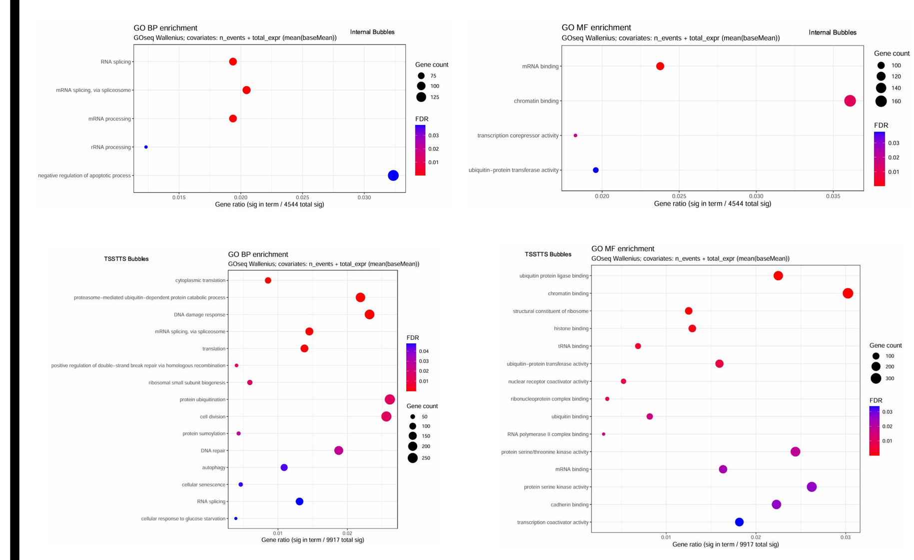


Figure 4. Precision–recall curves comparing performance of bipartition statistical models. Overall, EBapprox and EBmap showed the highest precision, while MLE and p-value combination strategies (fishercomb and mincomb) sometimes achieved higher recall but with reduced precision.

APPLICATION & RESULTS

- We applied the GrASE statistical models to detect differential alternative splicing between B cells and CD8 T cells. The RNA-seq data were obtained from Schmiedel et al. (Cell, 2018) and included 106 B-cell samples and 102 CD8 T-cell samples. The libraries are single-end and strand-specific with reverse-strand orientation.
- For internal bubbles, the four statistical models produced broadly similar numbers of significant events, but the empirical Bayes models were slightly more conservative than the non-EB approaches.
  - **EBapprox** identified 16,288 significant events out of 105,193 tested events, corresponding to **15.5%**.
  - **EBmap** identified 16,225 significant events out of 105,193 tested events, corresponding to **15.4%**.
  - **Wilcoxon Test** identified 17,197 significant events out of 105,193 tested events, corresponding to **16.3%**.
  - **Beta-binomial MLE** identified 16,356 significant events out of 96,349 tested events, corresponding to **17.0%**.
- Overall, EBapprox and EBmap produced the lowest proportions of significant calls among the tested models. This pattern is consistent with the simulation results, where empirical Bayes moderation improved false-positive control. A similar pattern was observed for TSS/TTS bubbles, where EBapprox and EBmap again produced the lowest proportions of significant calls among the tested models, at **16.9%** and **16.8%**, respectively.



Figures 5–8. GO Biological Process and Molecular Function enrichment dot plots of EBapprox significant genes in internal and TSS/TTS bubbles. Functional enrichment analysis was performed using over-representation analysis (ORA) separately for internal and TSS/TTS bubbles. Target gene sets were defined as the unique genes associated with at least one significant EBapprox event, while background gene sets were defined as all unique genes tested by GrASE, including genes associated with both significant and nonsignificant events. Enrichment was evaluated separately for GO Biological Process and Molecular Function terms. The analysis revealed enrichment of terms related to mRNA processing, RNA splicing, mRNA binding, and other gene-regulatory processes.

CONCLUSIONS

GrASE bipartitioning enables differential splicing analysis by comparing shared and distinct exonic parts within internal and TSS/TTS bubbles. In both simulation benchmarking and B-cell versus CD8 T-cell analysis, empirical Bayes models improved false-positive control and produced more conservative significant event sets. EBapprox significant genes were enriched for GO terms related to RNA splicing, mRNA processing, mRNA binding, and other gene-regulatory processes. Together, these results support GrASE as a robust framework for detecting biologically relevant differential alternative splicing events.

REFERENCES

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